

Smart Kanban

An AI-powered Kanban board built with **React, Express, Tailwind CSS, and Google Gemini AI** that helps users organize tasks, prioritize work, and improve productivity through intelligent task analysis.



Features

Task Management

- Create, edit, and delete tasks
- Drag & drop tasks across columns
- Priority labels (High, Medium, Low)
- Search tasks instantly
- Filter tasks by priority
- Local storage persistence
- Responsive user interface

AI Workspace

-  Suggest task priority
-  Break complex tasks into actionable steps
-  Generate board summaries
-  Productivity coaching
-  AI-powered Kanban chat assistant

Dashboard

- Task statistics
 - Progress overview
 - Clean and modern UI
 - Responsive layout
-

Tech Stack

Frontend

- React
- Vite

- Tailwind CSS
- JavaScript (ES6+)

Backend

- Node.js
- Express.js

AI

- Google Gemini API

Deployment

- Frontend: Vercel
- Backend: Render

Project Structure

```
smart-kanban
├── public/
├── server/
│   └── server.js
├── src/
│   ├── assets/
│   ├── components/
│   ├── services/
│   ├── App.jsx
│   └── main.jsx
├── package.json
├── tailwind.config.js
└── README.md
```

AI Features

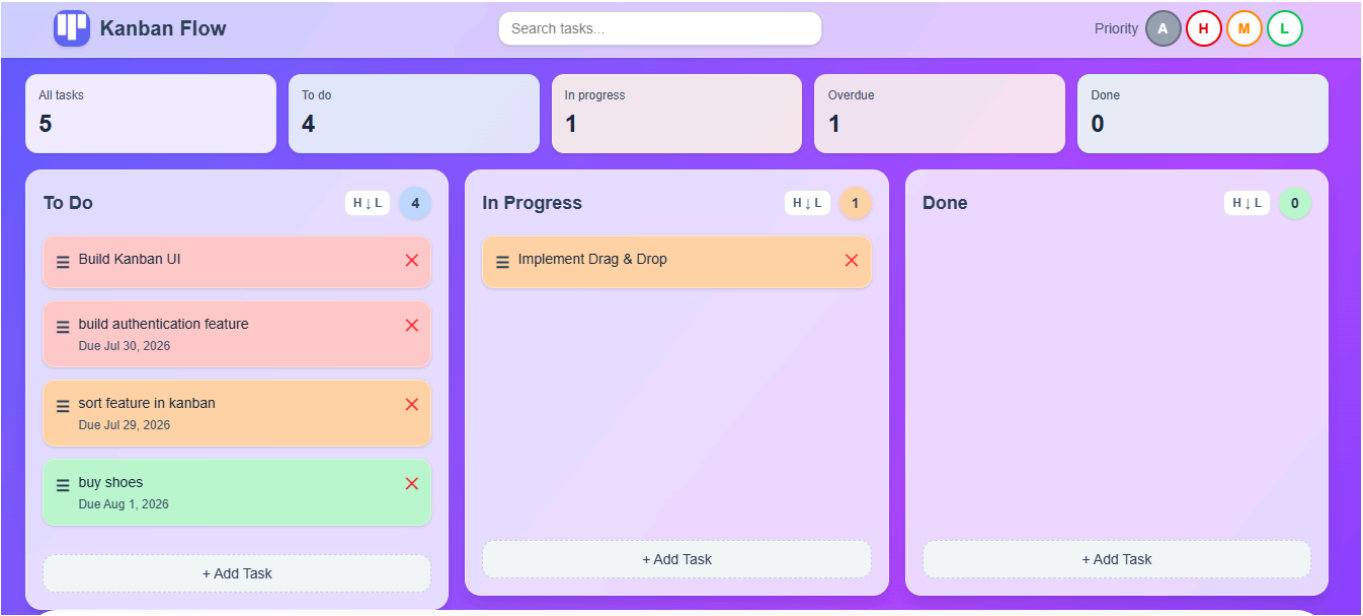
The AI assistant understands your current Kanban board and can:

- Analyze task priority
- Break tasks into implementation steps
- Summarize project progress
- Identify risks
- Suggest next actions
- Answer project-related questions
- Provide productivity guidance

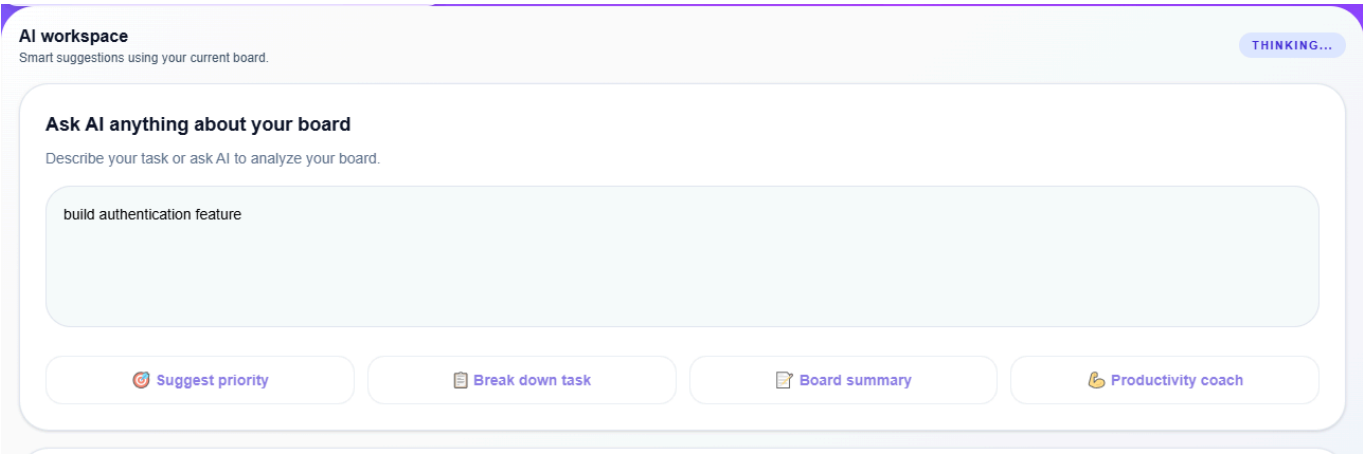
The backend securely communicates with the Google Gemini API, ensuring the API key is never exposed to the client.

Screenshots

Dashboard



AI Workspace



AI Chat Assistant

Clone the repository

```
git clone https://github.com/KeerthyHQ/smart-kanban.git
cd smart-kanban
```

Install dependencies

```
npm install
```

Start the frontend

```
npm run dev
```

Start the backend

```
npm run server
```



Environment Variables

Create a `.env` file inside the project root.

```
PORT=3001
GEMINI_API_KEY=YOUR_API_KEY
GEMINI_MODEL=gemini-flash-latest
```



How It Works

```
graph TD
    A[React Frontend] --> B[Express Backend]
    B --> C[Google Gemini API]
```

The frontend sends AI requests to the Express backend.

The backend securely calls the Gemini API and returns AI-generated responses.



Future Enhancements

- User Authentication
 - Cloud Database
 - Team Collaboration
 - Due Dates & Reminders
 - Activity History
 - Dark Mode
 - Analytics Dashboard
 - Export & Import Boards
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Key Concepts Practiced

- React Components
 - React Hooks
 - State Management
 - Drag & Drop UI
 - REST APIs
 - Express.js
 - AI Integration
 - Prompt Engineering
 - Environment Variables
 - Responsive Design
 - Local Storage
 - Component Architecture
-



Author

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Support

If you found this project useful, consider giving it a ★ on GitHub.

rt & Import Boards



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